STAT 801 — HOMEWORK 2

FALL 2016

DUE 4:30 PM, SEPT 19

Instructions for HW 2:

- **Report z-scores to the nearest 2 decimal places.**
- **Report final probability as a proportion (NOT a percent) to the nearest 4 decimal places.**

1. **Notation.** Suppose that a random variable, X, follows a Normal distribution. Specifically, $X \sim \mathcal{N}(-2, 16)$.
   (a) What is the value of the mean?
   (b) What is the value of the median?
   (c) What is the value of the mode?
   (d) What is the value of the standard deviation?
   (e) What is the value of the variance?
   (f) What is the total area under the density curve?
   (g) Multiple Choice. What is the range of values the random variable, X, can take on? (Enter the roman numeral corresponding to the correct answer, i.e. enter i, ii, iii, iv, or v.)
      i. -14 to 10
      ii. $-\infty$ to ∞
      iii. -50 to 46
      iv. -26 to 22
      v. -98 to 94

2. **Corn.** Yield of corn (bushels per acre) in Nebraska counties in 2009 can be described by a Normal distribution with a mean of 178 bushels per acre and a standard deviation of 16 bushels per acre. Use **the 68-95-99.7 Rule (Empirical Rule)** to answer the following questions. (Hint: Draw pictures of the normal curve.)
   (a) The middle 68% of counties have a corn yield **between** what two values?
   (b) What is the value of the 84^{th} percentile of corn yield for counties in Iowa?
   (c) What proportion of counties have a corn yield **between** 162 and 226 bushels per acre?
   (d) 0.15% of counties have a corn yield **less than** or equal to what value?
   (e) What proportion of counties have a corn yield of **at least** 210 bushels per acre?

3. **Gas Mileage**. For a particular model of car, the miles per gallon (mpg) rating obtained for highway driving follows a Normal distribution with mean 28.5 mpg and standard deviation 1.4 mpg. Use this information to answer the following questions.

   (a) What is the probability a car will obtain a miles per gallon rating for highway driving greater than 32 mpg? Answer this question by completing parts (i) and (ii).

      i. Provide the z-score corresponding to the above statement.

      ii. Based on your answer in (i), what is the probability a car will obtain a miles per gallon rating for highway driving greater than 32 mpg?

   (b) What is the probability a car will obtain a miles per gallon for highway driving less than 24 mpg? Answer this question by completing parts (i) and (ii).

      i. Provide the z-score corresponding to the above statement.

      ii. Based on your answer in (i), what is the probability a car will obtain a miles per gallon for highway driving less than 24 mpg?

   (c) Find Q_1 (First Quartile). Answer this question by completing parts (i) and (ii).

      i. Provide the z-score corresponding to the above statement.

      ii. Based on your answer in (i), provide the value of Q_1. (Round answer to 3 decimal places.)

   (d) Find Q_3 (Third Quartile). Answer this question by completing parts (i) and (ii).

      i. Provide the z-score corresponding to the above statement.

      ii. Based on your answer in (i), provide the value of Q_3. (Round answer to 3 decimal places.)

   (e) What is the value of the IQR for the distribution of miles per gallon ratings for highway driving? (Round answer to 3 decimal places.)

   (f) What is the mpg rating such that only 4% of all cars have an mpg rating larger than that mpg value? Answer this question by completing parts (i) and (ii).

      i. Provide the z-score corresponding to the above statement.

      ii. Based on your answer in (i), what is the mpg rating such that only 4% of all cars have an mpg rating larger than that mpg value? (Round answer to 2 decimal places.)

      iii. What percentile does the mpg rating obtained in part (ii) correspond to? (Enter value as a whole number, not a proportion.)

   (g) What proportion of cars will have an mpg rating within 3 mpg of the mean? Answer this question by completing parts (i) through (iii).

      i. Provide the SMALLER z-score corresponding to the above statement.

      ii. Provide the LARGER z-score corresponding to the above statement.

      iii. Based on your answers in (i) and (ii), what proportion of cars will have an mpg rating within 3 mpg of the mean?

   (h) Which car is more unusual, a car with a 32 mpg rating for highway driving or a car with a 25 mpg rating for highway driving? Explain your answer.

   (i) Let's say you bought a car of the particular model discussed in this question and extensive testing shows that you get, on average, 28.5 mpg for your car on the highway. Provide an interpretation of your car's mpg rating, i.e. what can you say about how your car compares to all other cars of this particular model?

4. **SAT Scores.** A professor at a small univeristy in New York surveys her introductory statistics students at the beginning of each semester. One of the questions she asks her students is what was the highest SAT score they received. The professor has collected data over many semesters and has SAT scores for 353 students. These data can be found on Blackboard (Assignments - Homework Assignments - Homework 2) in SAT.jmp/SAT.csv/SAT.sas/SAT.xlsx. Use your favorite software to create a histogram and a boxplot of the SAT scores as well as to calculate summary statistics. You do not need to give me the output, but you will need it to answer the following questions.

   (a) Based on the output, describe the overall shape of the distribution of SAT scores for introductory statistics students that attended this small university in New York.

   (b) Suppose you decide to use a Normal model to describe SAT scores for all introductory statistics students at this small university in New York. What possible values of the parameters (μ and σ) could you use for this Normal model?

   (c) The worst score a person can get on the SAT is a 400 and the best score a person can get is a 1600. A Normal model allows for any value, including values less than 400 and greater than 1600. Explain why it is still okay to use a Normal model to approximate the distribution of SAT scores.